

Milestone Project Plan (12-Week Semester)

Project phases.

Phase	Timeline	Deliverables	Target Assessment	Weight
Phase 1: Conceptual Design	Weeks 3 – 4	Problem statement, Business Rules, and a comprehensive ER/EER Diagram .	(Design)	20%
Phase 2: Logical Design & Normalization	Weeks 5 – 8	Relational Schema Mapping, Foreign Key definitions, and Normalization proofs (3NF/BCNF) .	(Analysis)	20%
Phase 3: Implementation & Queries	Weeks 9 – 10	SQL DDL Script, mock data insertion (DML), 10+ complex queries (Joins, Subqueries, Aggregations), and 2 Triggers/Views.	(Implementation)	30%
Phase 4: Integration & Defense	Weeks 11 – 12	Simple UI/Backend integration (Python/Flask, Node.js, or Java), Live Group Presentation, and Peer Evaluation .	(Teamwork/Comm)	30%

Evaluation

This is a 3-student teamwork project; NO "one student did all the work", please.

Group Score (70% of Project Grade)

- **Data Modeling Accuracy (25%):** Does the ER/EER capture all business rules? Is the relational mapping mathematically correct according to Elmasri's rules?

- **Normalization & Optimization (15%):** Are redundancies minimized? Are the tables strictly in 3NF/BCNF with sound justification?
- **SQL Technical Rigor (20%):** Are data types chosen optimally? Do the triggers, indexes, and constraints work without crashing?
- **System Functionality (10%):** Does the basic application front-end actually talk to the database successfully during the live demo?

Individual Score (30% of Project Grade)

- **Individual SQL Defense (15%):** During the live presentation, each student is asked to write or explain a specific complex query or modification on the fly.
- **Peer Evaluation (15%):** The individual's share of the group grade will be evaluated.